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Bridging the gap: Accelerating Open-source Agility with Cisco Silicon One Platforms

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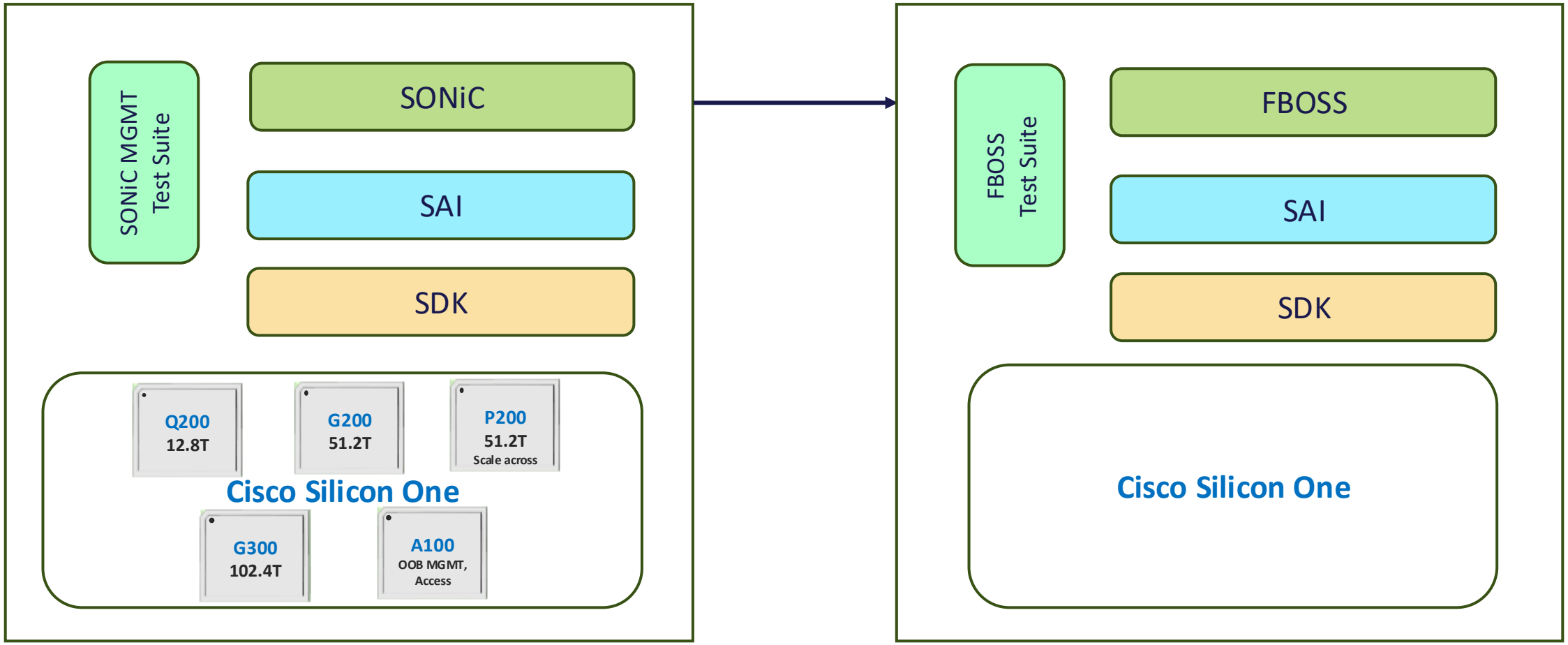
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Introduction

- Cisco Silicon One provides a consistent hardware foundation for scalable, disaggregated networking
- A common BSP, use of the SAI abstraction layer, and reusable validation workflow accelerate onboarding of open-source NOS solutions such as SONiC and FBOSS.
- **Agenda**
  - Open-source NOS architecture and ecosystem reuse
  - Integration Hurdles, Workflow and Benefits
  - Three-phase validation strategy and test workflow
  - Test coverage reuse, gaps, and integration challenges
  - Community collaboration and call to action

Open-NOS Architecture Flow



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Open-NOS Approach for Switch ASIC Testing

Area	SONiC sonic-mgmt	FBOSS Agent HW and SAI test	FBOSS Link Test
Main Purpose	Test SONiC features across T0/T1/T2 switch topologies	Test FBOSS agent features such as routing, ACLs, L2 learning, QoS, CoPP	Test end-to-end link and specific features such as PTP across cables
Test style and runner	Mostly Python tests with pytest; Ansible handles testbed setup and is used by tests to interact with devices	C++ Google Test cases built into agent hardware-test binary, run through run_test.py	Also Google Test in a FBOSS binary, run through run_test.py relying on qsfp_service for module programming
Typical Setup	Testbed with DUT, neighbor devices or simulated neighbors and packet injection/capture – generally with PTF	Standalone switch; no external cable connections are required. Data-plane tests inject packets through ports in loopback.	Physical cabled topology and generated Agent and QSFP configurations
What it exercises	Broad system and feature behavior; for example, routing, protocols, platform behavior and link-flap cases	Agent and ASIC feature behavior, including forwarding and warmboot checks	Interaction of wedge_agent, qsfp_service and FSDB, including link state, flaps, LLDP/fabric reachability, L1 diagnostics, PRBS and warmboot
Problem Solved	Does SONiC behave correctly in a networked testbed?	Does FBOSS operate this switch's ASIC dataplane features correctly with the desired behavior?	Do the ASIC, optics, ports and FBOSS services work across physical ports with expected link performance and timing?
Switch Requirement	**Uses SAI**		

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Integration Hurdles and Solution

Different Hardware

Use the same hardware:
- CPU: SKU, memory, BIOS, Kernel, Linux, Repo Libraries
- Modules – optics/active/passive cables and hardware/firmware versions
- Hardware Configuration
- Topology and peers

Different Test Suites

- Run same test suites including revisions
- Run with same test configurations

Different Software Versions

Use same software versions:
- Use same platform driver versions
- Use same SAI versions
- Use same switch SDK versions

Mixes can happen

Different Operational Semantics And Conditions

Run same network topologies, equipment, timing and conditions

Strategic design and testing

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Open-NOS Testing Workflow and Benefits

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NOS Developer End User

Cisco
Silicon One Provider

ODM(s)

Cisco Workflow with Silicon One Based Platforms

① Build FBOSS/SONiC in standardized container

② Link with Silicon One SAI libraries

③ Run Validation Tests on Silicon One Platforms

④ Run Feature tests on multi-node topology

⑤ Run Solution tests emulating Prod topology

⑥ Deliver validated SAI/SDK code to end user

⑦ Debug with Silicon One gRPC-based CLI and Python shell

Multiparty Coverage

- Hardware Validation
- SAI/SDK Validation
- Optics, Cables and Peer Interop

One Workflow

- Catches bugs earlier
- Easy Debug and Repro

Shared and common platform configs across vendors

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Three Phase Network Validation

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Phase	Scope	Purpose
Component-Level (UT, BSP, SAI/SDK)	Components validated in isolation — BSP on standalone HW, SAI/SDK in SIM,…	Ensure each building block works before integration
Per-Box Qualification (Dev Qual, NRT)	Complete device validated against intended network role	Validate functionality, performance, interop at box level
Solution-Level (NRT, System Test)	Network-wide role verification with multi-node topologies at datacenter scale	Confirm production readiness

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Test Surface Reuse Per Layer

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Layer	Test Surface	Reuse Level
BSP / Platform HW	Optics, sensors, fans, PSU, EEPROM, I2C, PCIe	HIGH — identical HW, identical tests
Data Plane (ASIC)	Forwarding, queuing, scheduling, ACL, QoS	HIGH — NOS-independent behavior
SAI / SDK Abstraction	SAI API → Cisco Silicon One SDK	HIGH — SAI vector tests directly reusable
Control Plane	FRR vs. FBOSS Agent + OpenR	MEDIUM — same topologies, NOS-specific scripts
Application / Management	CLI/gNMI vs. CLI/Thrift	LOW — FBOSS-specific tests needed

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NOS Qualification via existing topologies

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Test Area	Reuse Level	Notes
SAI/SDK Validation	FULL	PTF, SAI Challenger, SDK regression; already underway
BSP / Platform	HIGH	Board-level HW tests, minimal adaptation
Data-Plane / Forwarding	HIGH	Leaf, Spine, B2B, snake topologies reused
Performance & Latency	FULL	Latency is an ASIC property; HW-level measurement
Optics	FULL	DOM read via FBOSS CLI/Thrift; same HW path
QoS / PFC / RDMA	HIGH	IXIA-driven; PFCWD needs FBOSS-specific validation
Manufacturing (SNT)	FULL	NOS-independent by design

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Challenges

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Security

- Open NOS vulnerability detection and mitigation
- Security patches upgrade cadence

Interoperability

- Interoperability issues with third-party optics, NIC and switch vendors

Troubleshooting

- Takes longer to debug, mitigate and resolve issues

Integration

- Tools/scripts integrations challenges

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Open-NOS Engagement Models

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- Evolving partnership models
  - Wedge400C partnership through **Silicon only**
    - Cisco Silicon One Q200 on Wedge400C
    - Deployed in production since 2021
  - Wedge800C partnership through **Silicon only**
    - Cisco Silicon One G202 on Wedge800c
    - 32x800G OSFP (10G-800G)
  - Cisco 8501 partnership through gray box
    - Cisco Silicon One G200
    - Cisco/Meta OCP Inspired System
  - Cisco 8223 developed by Cisco
    - Cisco Silicon One P200
    - Cisco-designed System
- Through partnerships with FBOSS and SONiC communities

Platform	BW	Cisco Silicon, SDK, and SAI	HW & BSP	NOS Choice
Wedge400C	12.8T	Silicon One Q200	ODM	FBOSS/SONiC
Wedge800C	25.6T	Silicon One G202	ODM	
Cisco 8501	51.2T	Silicon One G200	Cisco	
Cisco 8223	51.2T	Silicon One P200	Cisco	

Wedge400C



Wedge800C



Cisco 8501



Cisco 8223



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Call to Action

- Problem to solve
  - Reduce the time and effort required to integrate new silicon, SDKs, and open-source NOS releases while maintaining feature parity, performance, and production quality.
- Call to Action:
  - Adopt standardized SAI, BSP, data-plane, and topology-based validation practices
  - Contribute test cases, fixes, and platform learnings to upstream SAI projects
  - Expand SONiC and FBOSS coverage for control-plane, telemetry, restart, and API validation
  - Share multi-vendor interoperability results and reusable test automation
  - Collaborate on consistent qualification criteria across silicon, platforms, and NOS implementations
- Available **today**: Cisco Silicon One platforms, common BSP/SDK integration, and existing validation workflows
  - Cisco 8501: https://www.opencompute.org/products/565/cisco-8501-sys-mt-64x800g-osfp-switch
  - Open Compute Project SAI: https://github.com/opencomputeproject/SAI
  - Cisco 8223: https://www.cisco.com/c/en/us/products/collateral/routers/8000-series-routers/8200-series-routers-ds.html
  - Cisco Silicon One: https://www.cisco.com/site/us/en/products/networking/silicon-one/index.html

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Thank you!



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